



2.4

Exponential Function Manipulation

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

| Teacher: Mohammad Abu-

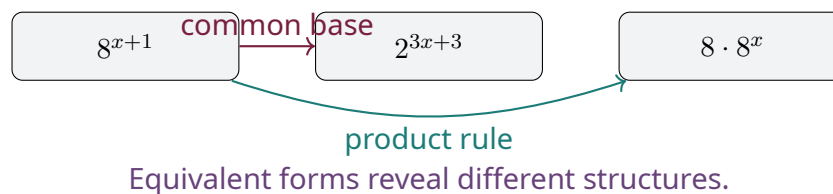
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Learning Targets

- Rewrite exponential expressions using exponent laws and common bases.
- Connect horizontal shifts to multiplicative output scaling.
- Interpret rational exponents and equivalent time scales.

Essential Question

How can equivalent exponential forms reveal different structural features?



Concept Framework

Key Idea

Exponent laws convert multiplication into addition of exponents and powers into multiplication of exponents.

Key Idea

A horizontal shift can be rewritten as an output scale: $b^{x-h} = b^{-h}b^x$.

Key Idea

Rational exponents represent roots and powers.

Key Idea

Changing time units changes the base but not the underlying model.

Formula Map**Formula and Structure**

$$b^{u+v} = b^u b^v$$

Formula and Structure

$$(b^u)^v = b^{uv}$$

Formula and Structure

$$b^{p/q} = \sqrt[q]{b^p}$$

Formula and Structure

$$ab^{x/h} = a(b^{1/h})^x$$

Worked Examples**Worked Example 1**

Problem. Rewrite 8^{x+1} with base 2.

Solution. $8^{x+1} = (2^3)^{x+1} = 2^{3x+3}$.

Worked Example 2

Problem. Rewrite $5 \cdot 3^{x-2}$ as $A \cdot 3^x$.

Solution. $A = 5/9$, so $(5/9)3^x$.

Worked Example 3

Problem. A quantity triples every 8 hours. Give an hourly factor.

Solution. $3^{1/8}$.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Rewrite or evaluate 4^{2x-1} .

2. Rewrite or evaluate $9^{x/2}$.

3. Rewrite or evaluate 5^{x+2} .

4. Rewrite or evaluate $16^{3/4}$.

5. Rewrite or evaluate $27^{2/3}$.

6. Rewrite $12 \cdot 4^{x-3}$ as $A \cdot 4^x$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.